

group can individually be simulated. Use 'Simulator' tab and the simulator window appears as shown in Fig. 11. Select the group to be simulated from the list. There are two methods of simulation:

1. Step simulation: Use '>>' and '<<' buttons to navigate, to step-up or down, respectively.

2. **Animated simulation:** This simu-

lation provides a real-world feeling and can run the group as LEDs really do.

Builder. Click on the 'Builder' tab (refer Fig. 12) to generate the assembly source file for AT89S51. Click 'Generate ASM' button to get the entire source code. Project.asm file appears in bin directory. Assemble the source program using suitable assembler and

burn the hex code into the microcontroller using a suitable programmer.

If you are trying to open an existing project, the files associated with the project are loaded, else you can start with a fresh project similarly as above.

EFY note. The source code of this article is available on www.efymag.com website.

PROJECT.ASM

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MOD51
;Program for Chaser Designer
ORG 0000H
LJMP MAIN

;Delay function.Put 16-bit delay in
milli seconds.R1-higher byte and R0-
lower byte.
DELAY:
MOV TLO,#0FFH
MOV TH0,#0EDH
SETB TR0
CONTINUE:JNB TF0,CONTINUE ;Delay for 1ms
CLR TF0
DJNZ R0,DELAY ;Lower
CJNE R1,#00H,LABEL ;Higher
RET
LABEL:
DEC R1
SJMP DELAY

;*****Main*****
MAIN:

;call subroutine FORWARD for 20 times
MOV R2,#014H
LABEL_0:
LCALL FORWARD
DJNZ R2,LABEL_0

;call subroutine DANCING for 10 times
MOV R2,#0aH
LABEL_1:
LCALL DANCING
DJNZ R2,LABEL_1

;call subroutine REVERSE for 20 times
MOV R2,#014H
LABEL_2:
LCALL REVERSE
DJNZ R2,LABEL_2

;call subroutine REVERSE for 10 times
MOV R2,#0aH
LABEL_3:
LCALL REVERSE
DJNZ R2,LABEL_3

LJMP MAIN

;
subroutine:
MOV P1,#080H
MOV R1,#01H
MOV R0,#091H
LCALL DELAY
MOV P1,#040H

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MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #020H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #010H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #008H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #04H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #02H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #01H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
RET

;DANCING subroutine
DANCING:
MOV P1, #0aaH
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #055H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
RET

;FORWARD subroutine
FORWARD:
MOV P1, #080H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #040H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #020H
MOV R1, #01H
MOV R0, #091H
LCALL DELAY
MOV P1, #010H
MOV R1, #01H
MOV R0, #091H

```

```

CALL DELAY
MOV P1,#008H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#040H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#020H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#010H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
RET

;REVERSE subroutine
REVERSE:
MOV P1,#010H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#020H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#040H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#080H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#010H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#020H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#040H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
MOV P1,#080H
MOV R1,#010H
MOV RO,#091H
LCALL DELAY
RET
END

```